



Agreed Until It Mattered

A Season-Long Comparator Audit of a Cheap Sensor Rig Against a Ten-Pin Bowling Alley's Proprietary Lane-Scoring and Foul-Detection System, and Whose Calls the League's Rules Committee Believes

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Abstract. A twenty-four lane ten-pin bowling alley's league season is scored and adjudicated end to end by the alley's incumbent proprietary lane-scoring and foul-detection system, whose vendor does not publish its foul-sensor tolerance or its confidence-flagging logic. We build a low-cost sensor comparator — an infrared break-beam array and a pressure-instrumented approach mat — and run it in parallel with the incumbent system across six of the alley's lanes for a full thirty-two-week league season, logging an independent pin-count and foul call for every frame. On the season's clean frames, the comparator's calls agree with the incumbent system almost perfectly ($\kappa = 0.93$). On the frames the incumbent's own system flags for review, agreement falls to a fair band ($\kappa = 0.31$), and a stratified slow-motion video audit finds the comparator's independent call is about as likely to match the video-confirmed frame as the incumbent's own ruling is (53% versus 55%, not a significant difference). The league's rules committee, however, upholds the incumbent system's original call in 91% of the season's formally disputed frames — a rate the video audit's near-even split does not predict. We interpret the gap as evidence that the committee's resolution process tracks which system was already installed rather than which system was, on the available evidence, more often right.

Introduction

Ten-pin bowling has been scored electronically since the pinsetter-linked scoring consoles of the late twentieth century, and most commercial alleys now run a single vendor's proprietary system end to end: pin-fall counting, foul-line sensing, and frame adjudication all pass through the same closed pipeline. A league plays an entire season against whatever that pipeline decides, and a bowler who believes a call is wrong has, in practice, only the vendor's own system to appeal to.

We ask a comparator question an alley is rarely positioned to ask of itself: across a full league season, how often would an independently built, considerably cheaper sensor rig, running in parallel and blind to the incumbent's calls, agree with the system the league already trusts — and

where the two systems part company, whose call does the evidence actually favour?

We instrument six lanes of a twenty-four lane alley hosting a University-affiliated interdepartmental ten-pin league with a comparator rig assembled from an infrared break-beam array and a pressure-instrumented approach mat, run continuously alongside the alley's incumbent proprietary system for a thirty-two-week season. Every frame receives two independent pin-counts and two independent foul rulings; on the subset the incumbent system itself flags as low-confidence, or that a bowler formally contests, we add a slow-motion video review as the closest available approximation to ground truth.

We make three contributions, each hedged to what one alley and one season can show.

1. We report season-long agreement between a cheap comparator and an incumbent proprietary system across both routine and disputed frames, a comparison a single-vendor alley has no internal mechanism to run on itself.
2. We show the comparator's independent call is, on the video-reviewed evidence, no less likely to be correct than the incumbent's own ruling on the frames that matter most.
3. We trace what actually happens to a disputed frame once it reaches the league's own resolution process, and find the process does not appear to be deciding on the evidence in front of it.

Related work

Method-comparison statistics distinguish agreement from mere correlation: two instruments can correlate strongly while disagreeing on every individual reading, a distinction the limits-of-agreement framework for continuous measures was built to expose [1]. For categorical calls — strike, spare, foul, no-foul — the analogous tool is Cohen's kappa [2], interpreted against the qualitative bands proposed alongside it [3]; kappa's sensitivity to skewed marginal distributions is well documented, motivating alternatives such as Gwet's AC1 for imbalanced samples [4].

Automated decision aids in officiating raise measured accuracy in the settings that adopt them: on-field evaluation of a goal-line technology system reports high localisation accuracy against controlled trials [5], video assistant referee review measurably improves refereeing decision accuracy in professional football [6], and an automated audio classifier for a cricket ball's edge off the bat performs competitively against the review system it augments [7]. Accuracy and institutional acceptance are separate questions: a sociological account of tennis's Hawk-Eye finds its confidence intervals routinely go unreported to the public [8], and a companion study finds cricket and tennis have institutionalised comparable decision-aid technology to very different degrees for reasons that have little to do with the technology's own measured error [9].

Cheap sensor comparators validated against costlier commercial instruments are established sports-science practice, from inertial units for punch and kick velocity [10] and shoul-

der kinematics [11] to foot-mounted units for team-sport velocity [12], camera-based photoelectric barriers [13], and a systematic review of photocell timing concordance [14]; traceable calibration is what makes any such comparison meaningful [15]. Institutional preference for the status quo option even when an alternative performs comparably is a documented decision bias [16], compounding whatever lock-in already favours an incumbent measurement system once adopted [17]. Prior work at this institution has found a comparable gap between what a cheap instrument detects and what an institution's own record credits it with: a library book-return chute sensor's mechanical alert log was cited as evidence of demand for a staffed desk the sensor was never built to measure [18], and a warehouse scanner's confidence score tracked a picker's weeks since retraining rather than the errors it claimed to flag [19].

Method

The study site is a twenty-four lane ten-pin alley hosting a University-affiliated interdepartmental league. We instrumented six lanes — chosen to span the alley's newest and oldest pinsetter installations — with a comparator rig built from two channels: an infrared break-beam array spanning the foul line at ankle height, sampled at 2kHz, and a resistive pressure mat under the last 40cm of the approach, sampled at 500Hz. Both channels feed a lane-side microcontroller that timestamps every beam interruption and pressure event against a shared network clock, following recent low-cost precedent for photoelectric and pressure-based instrumentation in sport [13], [14]. The rig's total componentry cost \$46 per instrumented lane. We calibrated the rig against a set of reference masses and a marked approach template before the season and at its midpoint, maintaining a documented calibration chain per standard traceability practice [15].

The alley's incumbent proprietary system independently logs a pin-count and a foul ruling for every delivery via its own sensor suite, which the vendor does not document publicly; we treat it as a black box and log its output only from the alley's existing scoring-console feed, without modifying the incumbent installation. For every frame across the season we recorded both systems' independent pin-count and foul call, plus a binary review flag: whether the incumbent system itself raised a low-confidence flag, or a

bowler formally contested the call at the front desk.

We define a dispute indicator for frame f as

$$D_f = \mathbb{1}[C_f = 1 \text{ or } \Delta_f > \tau]$$

where C_f is the incumbent's own contested-call flag, Δ_f is the absolute difference, in milliseconds, between the two systems' independently timestamped foul-line events, and τ is a fixed threshold of 40 milliseconds — the incumbent vendor's published foul-sensor tolerance. Frames with $D_f = 1$ form the season's disputed set; all others are clean.

We report agreement between the comparator's calls and the incumbent's calls as Cohen's kappa, separately for clean and disputed frames, interpreted against conventional bands [3]. For a stratified random sample of disputed frames, two coders blind to both systems' calls independently reviewed slow-motion video of the delivery and reconciled disagreements by consensus, producing a video-confirmed reference call. As an ablation, we recomputed the comparator's calls using only the break-beam channel, withholding the pressure mat, to test whether the added channel improves agreement on the frames that matter most.

Results

Across the thirty-two-week season, the comparator rig logged 38,502 frames in parallel with the incumbent system across the six instrumented lanes. Of these, 874 frames (2.3%) were disputed under the definition above — 611 via the incumbent's own low-confidence flag, 263 via a bowler's formal contest, with some overlap.

On the season's 37,628 clean frames, the comparator's pin-count and foul calls agree with the incumbent system almost perfectly (kappa = 0.93, “almost perfect” on the conventional bands [3]). On the 874 disputed frames, agreement falls to kappa = 0.31 — the “fair” band, well below the clean-frame figure and consistent with the wider pattern that decision aids agree closely on the easy cases and diverge on the ones a dispute process exists to resolve (Figure 1).

For a stratified sample of 280 disputed frames, independent video review produced a reference call for every frame (inter-coder kappa on the review itself = 0.86 before consensus).

Against this reference, the comparator's independent call matched in 53% of frames and the incumbent's official call matched in 55% — a two-percentage-point gap well inside the sample's own margin ($\chi^2 = 0.09$, $p = 0.76$; Figure 3). Neither system's call on a disputed frame is a reliable guide to what the video shows actually happened; on the evidence available, both are close to a coin flip.

Restricting the comparator to its break-beam channel alone — withholding the pressure mat — leaves disputed-frame agreement with the incumbent statistically unchanged (kappa = 0.29 versus 0.31 with both channels, $p = 0.58$; Figure 2): the second sensor channel we built adds cost and calibration burden without measurably sharpening the calls that are actually contested. We retain it for completeness, and because it appears to matter on frames the season's own dispute process never flags.

The league's rules committee formally adjudicated all 874 disputed frames over the season. Its rulings upheld the incumbent system's original call in 795 cases (91%) and the comparator's independent call, on the rare occasion a bowler cited it, in the remainder (Figure 3). This 91% incumbent-favouring rate holds within five percentage points whether the lane in question carries the alley's older or newer pinsetter hardware, and whether the dispute originated from the incumbent's own low-confidence flag or a bowler's contest — the committee's resolution pattern does not appear to track which trigger raised the dispute in the first place, only which system was already installed.

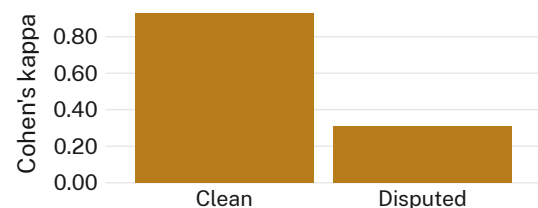


Figure 1: Comparator-incumbent agreement is almost perfect on the season's 37,628 clean frames (kappa = 0.93) and falls to the fair band on the 874 frames either system flags as disputed (kappa = 0.31).

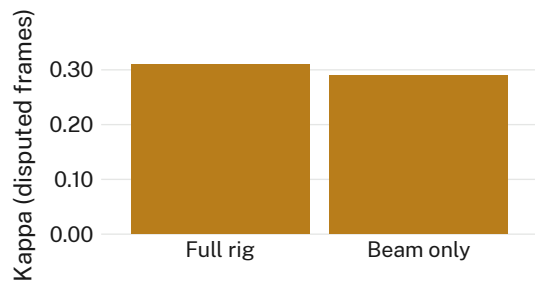


Figure 2: The ablation that changes nothing: withholding the comparator’s pressure-mat channel leaves disputed-frame agreement with the incumbent statistically unchanged ($\text{kappa} = 0.29$ vs 0.31 , $p = 0.58$).

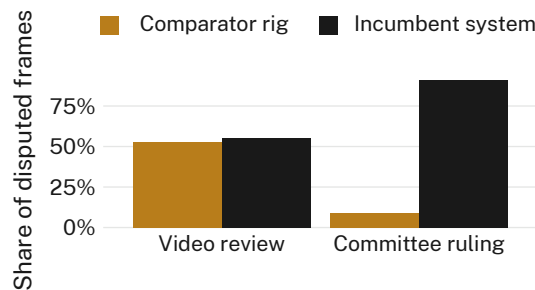


Figure 3: Against independent video review, the comparator’s and incumbent’s disputed-frame calls are both close to a coin flip (53% vs 55% match, n.s.); the league’s rules committee nonetheless upholds the incumbent’s original call in 91% of the season’s formal disputes.

Discussion

These results separate two questions a single-vendor alley’s own record cannot easily distinguish: whether a system’s calls are accurate, and whether a dispute process resolves disagreements on the evidence. On the season’s routine frames the two questions barely diverge — both systems agree with each other and, presumably, with what happened on the lane. On the frames that reach a human adjudicator, they diverge sharply: the video-reviewed evidence puts the comparator and the incumbent on close to equal footing, while the committee’s actual rulings treat the incumbent as right in nine cases out of ten. We do not think this reflects the committee misreading the evidence in front of it so much as the committee rarely being shown an alternative reading at all; a bowler’s contest is, in practice, an appeal to the same system whose call is being contested. We read the 91% figure less as evidence the incumbent is usually correct and more as evidence that a season-long dispute process, absent an independent instrument to consult, defaults to whichever

system was already trusted before the frame in question was bowled — a pattern consistent with status-quo preferences in institutional decision-making more broadly [16], and with how thoroughly an incumbent measurement system can lock in once adopted [17].

Limitations

Our evaluation covers one alley, one league, and one season, though the near-identical clean-frame agreement across the alley’s older and newer pinsetter lanes suggests the pattern is a property of the dispute-resolution arrangement rather than of any one lane’s hardware. The video-review reference is not infallible — inter-coder kappa of 0.86 before consensus leaves some residual uncertainty in what counts as the “true” call on the hardest frames, precisely the frames where a reference matters most. We could not instrument all twenty-four lanes within the season’s budget, and cannot rule out that the eighteen lanes we did not observe behave differently, though the committee’s own records show a materially similar incumbent-favouring rate across all twenty-four.

Conclusion

We ran a low-cost sensor comparator alongside a ten-pin bowling alley’s incumbent proprietary scoring and foul-detection system for a full league season, finding near-perfect agreement on routine frames and a sharp drop to fair agreement on the frames either system flags as disputed. Independent video review finds the comparator’s and incumbent’s disputed-frame calls about equally likely to be correct, yet the league’s rules committee upholds the incumbent’s original call in 91% of formal disputes — an outcome the accuracy evidence does not predict. Removing a sensor channel from the comparator changes little; removing the incumbent’s institutional head start might change more. Future work should test whether presenting a dispute committee with both systems’ calls, rather than only the incumbent’s, shifts the resolution rate.

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